

# South Australian Electricity Market

Capture Price Dynamics, Technology Economics & Portfolio Recommendation · Jul 2022 – Jun 2024

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## INVESTMENT THESIS

SA is a structurally bifurcated market — dispatch flexibility and timing earn premium revenue while raw generation volume does not. The optimal portfolio is a 60/40 wind-plus-storage hybrid, supplemented by selective solar only where firm contracts or co-located storage protect revenue.

### 01 · MARKET CONTEXT

#### Price trajectory & generation mix

**\$101**

2yr avg spot price /MWh

**\$375**

Jul 2022 gas-shock spike

**56.6%**

2yr avg renewable share

**-22%**

Grid gen peak-to-peak decline

#### SPOT PRICE DRIVERS

- Jul 2022 gas shock (Russia-Ukraine) **\$375/MWh**
- May-Jun 2023 wind drought spike **\$160-195**
- Stable post-shock corridor **\$60-80**

#### INVESTOR TAKEAWAY

When renewables fail, dispatchable assets capture concentrated revenue. A renewable-only portfolio is structurally short scarcity premium.

SA system security direction costs rose from \$7.4M (Q2 2022) to \$21.7M (Q2 2023). Rooftop PV displaced 22% of grid demand peak-to-peak.

### 02 · TECHNOLOGY ECONOMICS

#### Capture price hierarchy — 17× spread across technologies

|         |  |        |
|---------|--|--------|
| Diesel  |  | ~\$430 |
| Battery |  | ~\$170 |
| Gas     |  | ~\$170 |
| Wind    |  | ~\$70  |
| Solar   |  | ~\$25  |

Fleet DWP reference: \$173/MWh · Dispatch-Weighted Price (DWP), Jul 2022–Jun 2024

#### CAPTURE FACTOR TREND (ROLLING 90-DAY)

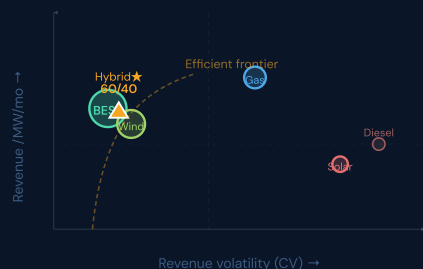
- Battery — peaked at 5.3× TWP, now compressing **↓ compress**
- Gas — surged from 1.7× to 4.9× TWP **↑ widening**
- Wind — positive trend +0.106/yr, rising above 1.0× **↑ +0.11**
- Solar — structural decline -0.026/yr (cannibalisation) **↓ -0.026**

#### SOLAR CANNIBALISATION

SA prices average ~\$0/MWh 11:00–14:00, peak at ~\$240/MWh at 18:00 — solar dispatches at exactly the wrong time. No operational fix within the technology itself.

### 03 · RISK-RETURN SYNTHESIS

#### Efficient frontier & portfolio recommendation



**\$18k**

Battery /MWh/mo · CV 0.55

**\$16k**

Wind /MWh/mo · CV 0.68

**\$17k**

60/40 hybrid · CV 0.55

#### PORTFOLIO RECOMMENDATIONS

##### HIGHEST CONVICTION

#### Storage — deploy now

Battery on efficient frontier. Early-mover premium is real but finite. Act within 24–36 months.

##### SCALE PLAY

#### Wind — deployable scale

Absorbs institutional capital. Asset-level selection critical — CLEMGPWF CF of 33.8% vs peer average.

##### AVOID

#### Standalone solar

Structural cannibalisation, low capture price. Only viable with PPA or co-located storage.

##### TACTICAL

#### Gas — exit-defined

High absolute return, high risk. Decarbonisation exposure. Tie exit to storage build-out milestones.

### CAPACITY FACTOR BENCHMARKS

**25–34%**

**10–20%**

**10–16%**

### RISK REGISTER

- Storage saturation — faster-than-modelled battery CF compression as SA storage queue commissions 2025–2027

Wind CF · benchmark 28–38%

Solar CF · benchmark 22–28%

Gas CF · peaking/firming role

CRITICAL FRAMING

The challenge facing SA renewables is **not operational underperformance** — assets hit resource targets. The problem is purely revenue: fix via contracting and storage co-location, not technology switching.

Battery net revenue modelled via rule-based arbitrage: charge bottom-12 daily intervals, discharge top-12, RTE = 0.85. Monthly battery revenue oscillates \$8k–\$42k/MW vs wind \$5k–\$25k and solar \$0–\$15k.

- **Policy reversal** — Federal CIS or state scheme changes reducing contracted revenue floors for new renewables
- **Connection delays** — transmission build and connection queue delays affecting deployment timelines and IRR
- **Gas price volatility** — domestic gas movements affecting peaker economics and low-VRE price formation
- **Resource scarcity events** — multi-month wind/solar droughts may be under-represented in the 2-year window

Sources: AEMO Quarterly Energy Dynamics Q2 2023, Q3 2025 · 2025 Electricity Statement of Opportunities · AEMO Generation Information Page · SA1 5-minute dispatch data